

I. Introduction

As a result of urbanization and industrialization, the issue of disposing waste has become significant. The volume of generated waste is now more than the capacity of existing mechanisms, such as manual separation and fixed plants. It is clear that traditional ways of disposing waste resulted in waste piling up to the brim, pollution and wastage of resources in recycling of valuable materials. For compensating for the impact on the environment, we must be smarter and faster in the way that we do things more efficiently and are more eco friendly in tackling the problem of wastage in the future.

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Authors

N. Krishnammal

Dept of Artificial Intelligence and Data Science, Vel Tech Rangarajan Dr. Sagunthala, R&D Institute of Science and Technology, Avadi, Chennai, India

J. Navadeep

Dept of Artificial Intelligence and Data Science, Vel Tech Rangarajan Dr. Sagunthala, R&D Institute of Science and Technology, Avadi, Chennai, India

B. Sai Sumanth

Dept of Artificial Intelligence and Data Science, Vel Tech Rangarajan Dr. Sagunthala, R&D Institute of Science and Technology, Avadi, Chennai, India

K. Priya

Dept of Artificial Intelligence and Data Science, Vel Tech Rangarajan Dr. Sagunthala, R&D Institute of Science and Technology, Avadi, Chennai, India

Figures



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A Hybrid Approach for Predictive Waste Classification and Resource Optimization using Deep Transfer Learning and Reinforcement Learning

Publisher: IEEE Cite This PDF

N. KrishnammalJ. NavadeepB. Sai SumanthK. PriyaAll Authors



Abstract	Abstract: The fast increasing quantity of waste produced in the world today is an environmental sustainability and resource efficiency crisis. Conventional waste management strategies are not working since they are inefficiently sorting and losing materials, with additional pollution, loss of resources, and additional environmental burdens. The objective of this study is to develop an AI model with Transfer Learning and Reinforcement Learning to address such issues. Transfer Learning improves the performance of waste classification in various classes in the scenario of limited labeled samples through the use of pre-trained deep
Document Sections	
I. Introduction	
II. Literature Review	
III. Methodology	

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Navadeep Jakkamsetti <navadeepjakkamsetti26@gmail.com>

IEEE I3TCON 2026- PUBLISHED ONLINE

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